



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester M.C.A. (CBCS) Degree Examinations
 August 2023

22MCA204 – DATA WAREHOUSING AND DATA MINING

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing One full question from each Unit.

		Unit – I			
		Marks	BT*	CO*	PO*
1.	a) List the differences between OLAP and OLTP.	5	L*1	1	1
	b) What are Data Marts? Brief on its types.	5	L1	1	1
	c) Elaborate the following:				
	i. Concept Hierarchy				
	ii. ETL Process	10	L2	1	1
2.	a) Compare Star Schema and Snowflake Schema used for designing multidimensional data model.	5	L2	1	1
	b) Define metadata. Explain the contents of metadata repository.	5	L2	1	1
	c) Brief on the following:				
	i. Bottom-up Approach for Data Warehouse development				
	ii. Apex cuboid				
	iii. Dice operation				
	iv. Distributive measures	10	L2	1	1
		Unit – II			
3.	a) What are missing values? Explain various strategies to deal with missing values.	4	L2	2	1
	b) Describe the following attribute types with suitable examples.				
	i. Ordinal attribute				
	ii. Quantitative attribute				
	iii. Continuous attribute				
	iv. Asymmetric attribute	6	L3	2	1
	c) Explain following with respect to Datawarehouse implementation:				
	i. Bitmap indexing				
	ii. Partial materialization				
	iii. Multidimensional OLAP (MOLAP) servers	10	L2	2	1
4.	a) For the following vectors x and y, calculate the indicated similarity or distance measures.				
	i. $x=(1,1,1,1)$, $y=(2,2,2,2)$: cosine, Euclidian				
	ii. $x=(0,1,0,1)$, $y=(1,0,1,0)$: cosine, Euclidian, Jaccard	5	L2	2	1
	b) What are datasets? Describe the general characteristics of data sets.	5	L2	2	1
	c) What is Data preprocessing? Explain following preprocessing techniques with suitable examples.				
	i. Aggregation				
	ii. Sampling	10	L2	2	1
		Unit – III			
5.	a) Define Association Rule. How do you measure the strength of an association rule? Illustrate.	5	L2	3	1
	b) Write the pseudocode for the frequent itemset generation of the Apriori algorithm.	5	L2	3	1
	c) Explain the process of FP tree construction with an example.	10	L2	3	1
6.	a) State and explain Apriori Principle for candidate itemset pruning.	5	L2	3	1

- b) Consider the following data set:

Customer ID	Transaction ID	Items Bought
1	0001	{a, d, e}
1	0024	{a, b, c, e}
2	0012	{a, b, d, e}
2	0031	{a, c, d, e}
3	0015	{b, c, e}
3	0022	{b, d, e}
4	0029	{c, d}
4	0040	{a, b, c}
5	0033	{a, d, e}
5	0038	{a, b, e}

- i) Compute the support for item sets {e}, {b, d}, and {b, d, e} by treating each transaction ID as a market basket.
- ii) Use the results in (i) to compute the confidence for the association rules {b, d} → {e} and {e} → {b, d}. Is confidence a symmetric measure?
- iii) Repeat (i) by treating each customer ID as a market basket. Each item should be treated as a binary variable (1 if an item appears in at least one transaction bought by the customer, and 0 otherwise).

5 L3 3 1

- c) Consider following transactions:

Transaction	Items
T1	I1, I2, I3
T2	I2, I3, I4
T3	I4, I5
T4	I1, I2, I4
T5	I1, I2, I3, I5
T6	I1, I2, I3, I4

Generate frequent item sets Using Apriori Algorithm. Also generate association rules from it. Given: Minimum support=3, Confidence= 60%.

10 L3 3 1

Unit – IV

7. a) Explain the following methods used to evaluate the performance of a classifier.
- Holdout method
 - Random subsampling
- b) Compare rule based and class based ordering schemes.
- c) Write sequential covering algorithm. Explain the same with an example.
8. a) Define Hunt's algorithm for building decision tree. Illustrate the same with suitable example.
- b) Write a short note on following:
- Confusion matrix
 - Mutually exclusive rules
 - Nearest neighbours

5 L2 4 1

5 L2 4 1

10 L2 4 1

10 L2 4 1

10 L2 4 1

Unit – V

9. a) Distinguish following types of clustering:
- Hierarchical and partitional clustering
 - Complete and partial clustering
 - Exclusive and overlapping clusters
- b) Explain following methods of defining proximity between clusters:
- Single link
 - Complete link
 - Group average
- c) Explain various classification-oriented measures of cluster validity.
10. a) Explain different types of clusters with suitable examples.
- b) Explain how basic K-means algorithm is used for clustering.

6 L2 5 1

6 L2 5 1

8 L2 5 1

10 L2 5 1

10 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
